THE MORPHISM FUNCTOR FROM A FINITE LOCALLY FREE SCHEME IS REPRESENTABLE BY A SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 20213-20254, immediately following the statement at 20200-20211.

Editorial note. One same-chapter lemma and one cross-chapter glueing lemma are recorded in assembly.external_references.

Editorial note. The statement's condition (2) is used exactly once, at the covering step inside the span.

Allowed standard prerequisites (editorial).

- Representability when the source is finite free over an affine base. chapter lemma lemma-hom-from-finite-locally-free-into-affine, cited in the proof. Provides the local model: over an affine piece of S the functor is representable by an affine scheme.
- Glueing functors. Schemes, lemma schemes-lemma-glue-functors, cited in the proof. Glues the local representing objects along the affine opens of S into a scheme representing the whole functor.
- Affineness of finitely many points in a fibre. hypothesis (2) of the statement, used at proof lines 20206-20208 and 20241-20247. Ensures that any finite set of points in a fibre of X over S lies in an affine open, which is exactly what the local representability needs to cover the functor.

Source reference. Original lemma 0BL3: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20171–20171. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01JJ: focused-seeds/geometry/S9-morph-hom-locally-free-representable/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $Z \to S$ and $X \to S$ be morphisms of schemes. Assume*

(1) *$Z \to S$ is finite locally free, and*
(2) *for all $(s, x_1, \ldots, x_d)$ where $s \in S$ and $x_1, \ldots, x_d \in X_s$ there exists an affine open $U \subset X$ with $x_1, \ldots, x_d \in U$.*

Then $Mor_S(Z, X)$ is representable by a scheme.

Proof. Consider the set I of pairs (U, V) where $U \subset X$ and $V \subset S$ are affine open and $U \to S$ factors through V. For $i \in I$ denote (U_i, V_i) the corresponding pair. Set $F_i = Mor_{V_i}(Z_{V_i}, U_i)$. It is immediate that F_i is a subfunctor of $Mor_S(Z, X)$. Then we claim that conditions (2)(a), (2)(b), (2)(c) of Schemes, Lemma 01JJ which proves the lemma.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Condition (2)(a) follows from Lemma 0BL3.

To check condition (2)(b) consider T/S and $b \in Mor_S(Z, X)(T)$. Thinking of b as a morphism $T \times_S Z \to X$ we find an open $b^{-1}(U_i) \subset T \times_S Z$. Clearly, $b \in F_i(T)$ if and only if $b^{-1}(U_i) = T \times_S Z$. Since the projection $p : T \times_S Z \to T$ is finite hence closed, the set $U_{i,b} \subset T$ of points $t \in T$ with $p^{-1}(\{t\}) \subset b^{-1}(U_i)$ is open. Then $f : T' \to T$ factors through $U_{i,b}$ if and only if $b \circ f \in F_i(T')$ and we are done checking (2)(b).

Finally, we check condition (2)(c) and this is where our condition on $X \to S$ is used. Namely, consider T/S and $b \in Mor_S(Z, X)(T)$. It suffices to prove that every $t \in T$ is contained in one of the opens $U_{i,b}$ defined in the previous paragraph. This is equivalent to the condition that $b(p^{-1}(\{t\})) \subset U_i$ for some i where $p : T \times_S Z \to T$ is the projection and $b : T \times_S Z \to X$ is the given morphism. Since p is finite, the set $b(p^{-1}(\{t\})) \subset X$ is finite and contained in the fibre of $X \to S$ over the image s of t in S. Thus our condition on $X \to S$ exactly shows a suitable pair exists. $\square$